SIMATS ENGINEERING

Assessment Tool 2 — Answer Key
Course: Artificial Intelligence (CSA17) | CO3 | Annexure A

Q1. Which of the following best describes a Logical Agent in AI?
Answer: b) An agent that uses a knowledge base and logical inference to decide actions

Q2. In Propositional Logic, '$P \rightarrow Q$' is logically equivalent to:
Answer: d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:
Answer: b) A conjunction of disjunctions (clauses)

Q4. In FOL, the sentence $\forall x$: Student(x) $\rightarrow$ Passes(x) means:
Answer: b) Every entity that is a student passes

Q5. Unification in FOL is the process of finding substitutions that:
Answer: b) Make two logical expressions syntactically identical

Q6. What is the core function of a logical agent in AI?
Answer: b) Using logic and knowledge representation to reason and make decisions

Q7. Which component of a logical agent is responsible for storing facts and rules about the world?
Answer: c) Knowledge Base (KB)

Q8. What does a knowledge-based agent do immediately after obtaining a percept from the environment?
Answer: c) Updates the Knowledge Base with the new percept

Q9. Which operation is used to add new information to a Knowledge Base?
Answer: b) TELL

Q10. What is a proposition in formal logic?
Answer: b) A declarative statement that is either true or false

Q11. Which of the following is NOT a proposition?
Answer: c) "Close the door."

Q12. Which logical operator is represented by the symbol $\neg$?
Answer: c) NOT

Q13. The AND ($\wedge$) operator is true only when:
Answer: b) Both propositions are true

Q14. A proposition that is always true, regardless of the truth values of its variables, is called a:
Answer: b) Tautology

Q15. In an implication ($P \rightarrow Q$), the statement is false only when:
Answer: b) P is true and Q is false

Q16. What is the number of rows in a truth table for a logical expression with 3 propositions?
Answer: c) 8

Q17. Which of these represents one of De Morgan's Laws?
Answer: a) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

Q18. A proposition that is sometimes true and sometimes false is known as a:

Answer: c) Contingency

Q19. Which logical connective is true if and only if both propositions have the same truth value?

Answer: b) Biconditional ($\leftrightarrow$)

Q20. According to operator precedence, which operator should be evaluated first?

Answer: c) NOT ($\neg$)

Q21. What is the process of deriving a conclusion from given premises called?

Answer: c) Inference

Q22. Which inference rule follows the form: If P, then Q; P; Therefore, Q?

Answer: b) Modus Ponens

Q23. Which valid reasoning pattern is represented by: $P \rightarrow Q$; $\neg Q$; Therefore, $\neg P$?

Answer: c) Modus Tollens

Q24. What can be concluded from the premises ($P \rightarrow Q$) and ($Q \rightarrow R$) using Hypothetical Syllogism?

Answer: c) $P \rightarrow R$

Q25. The rule that states if "P or Q" is true and "Not P" is true, then "Q" must be true is called:

Answer: c) Disjunctive Syllogism

Q26. Which rule allows you to conclude "P" from the premise "P and Q"?

Answer: b) Simplification

Q27. The inference rule Resolution follows which symbolic form?

Answer: c) $P \vee Q, \neg P \vee R \therefore Q \vee R$

Q28. Which of the following is an example of the "Addition" rule?

Answer: b) P, therefore P or Q

Q29. "If it rains, the ground is wet. The ground is wet. Therefore, it rained." This is an example of which invalid reasoning pattern?

Answer: b) Affirming the Consequent

Q30. What is the name of the invalid pattern: $P \rightarrow Q$; $\neg P$; Therefore, $\neg Q$?

Answer: b) Denying the Antecedent